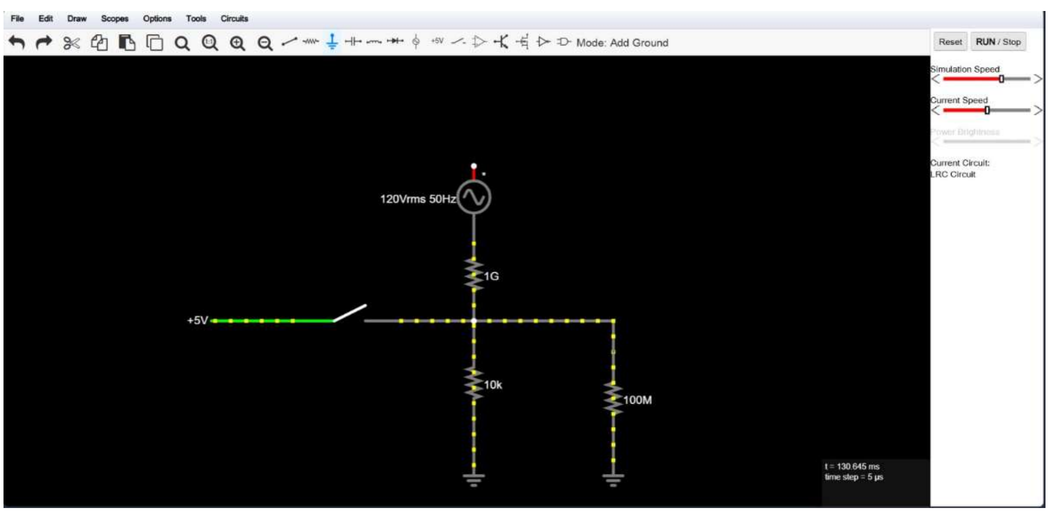
PCB & Circuits

❖What is Crosstalk in PCB and How to reduce it?

Crosstalk is unwanted electromagnetic coupling between nearby signal traces. A changing signal on one trace (aggressor) induces noise on another trace (victim). It can happen because of Traces routed too close together ,Long parallel traces ,Fast switching signals (high dV/dt) and Poor ground return path.

To reduce it: Increase spacing between traces. Keep parallel routing as short as possible, Use differential pairs for high-speed signals, Add ground guard traces, Slow edge rates if possible and Route adjacent layers perpendicular to each other.

❖Circuit



- **What is this circuit's primary function?**

  It creates stable digital input and prevents the input from floating.

  - Switch is open so input LOW
  - Switch is closed so input HIGH

  It prevents the input from floating.

- **What do the components inside the dashed box represent? Explain in detail.**

  It represent **environmental electrical noise**.

  Specifically:

  - 50 Hz AC source which is mains electricity
  - R3 (1 GΩ) which is weak capacitive coupling between surrounding wires and the circuit

  Even without touching the circuit, nearby AC wiring produces an electric field that induces a tiny current.

- **R1 (the main character)**

  1) What happens if you remove $R1$? (Simulate the result and explain).

  The input becomes floating.
  Simulation will shows: random HIGH/LOW readings and unstable voltage .

2)What is an input called when a resistor like R1 is missing?

Floating Input.

3)How should the resistance value for R1 be selected in a real circuit?

R1 should be low enough to firmly pull the pin LOW and be high enough to avoid wasting current when the switch is pressed

- ## What is the arrow likely pointing to, and what role does $R2$ serve? What could we call $R2$?

Points toward the microcontroller input pin.

R2 models the very high input resistance of the microcontroller input.

- ## If our active signal voltage were 0V instead of 5V, how should the circuit be modified? What is this new circuit topology called?

Instead of pulling the input LOW using R1,
Use:

- resistor from input to +5V
- switch connects input to GND

Which is called a Pull-Up Resistor configuration.

- ## What would happen if you brought your hands close to (or touched) an unshielded circuit like this?

  It picks up the surrounding 50 Hz mains electric field, and floating input receives this noise and may randomly change state, oscillate or trigger false button presses .This is exactly why pull-up and pull-down resistors are necessary.

❖Explain the difference between Decoupling Capacitors and Bulk Capacitors and when we use them.

Decoupling Capacitors

Purpose:

- Remove high-frequency noise
- Stabilize IC supply voltage

Placed:

- Directly beside every IC VCC pin.

Bulk Capacitors

Purpose:

- Store energy
- Prevent voltage sag
- Smooth power supply ripple

Placed:

- Near voltage regulators
- At power entry
- Near motors and high-current loads

❖What are differential pairs in PCB? Detail the critical rules for routing them.

A differential pair consists of two traces carrying equal and opposite signal for example USB ,Ethernet ,HDMI and PCIe and the advantages of it is

- Better noise immunity
- Lower EMI
- Higher signal integrity

The Routing Rules

- Keep traces the same length.
- Maintain constant spacing.
- Route together.
- Avoid sharp bends.
- Match impedance.
- Keep return path continuous.

❖In PCB design, there are various techniques for connecting components, one of them is polygon pours. What are polygon pours and why is it preferred in Power and GND connections?

A polygon pour is a large copper area automatically filled on a PCB, and it is commonly used for Ground and Power planes .

Advantages:

- Lower resistance
- Better grounding
- Reduced voltage drop
- Easier routing
- Lower EMI
- Better return current paths

❖ **You have an analog sensor signal with an output range of 0V to 12V, but your microcontroller's analog input pin can only handle a maximum of 5V. Worse yet, you are completely broke, so you need the absolute cheapest solution to step down this voltage range to safely read the signal with your ADC without losing signal information.**

Input: 0 - 12V
ADC: 0 - 5V
Budget: 6mW

$$I = \frac{P}{V} = \frac{6mW}{12V} = 0.5mA$$

$$R_T = \frac{12V}{0.5mA} = 24K\Omega$$

$\therefore V_{out} = 5V$

$$\frac{R_2}{R_1 + R_2} = \frac{5}{12}$$

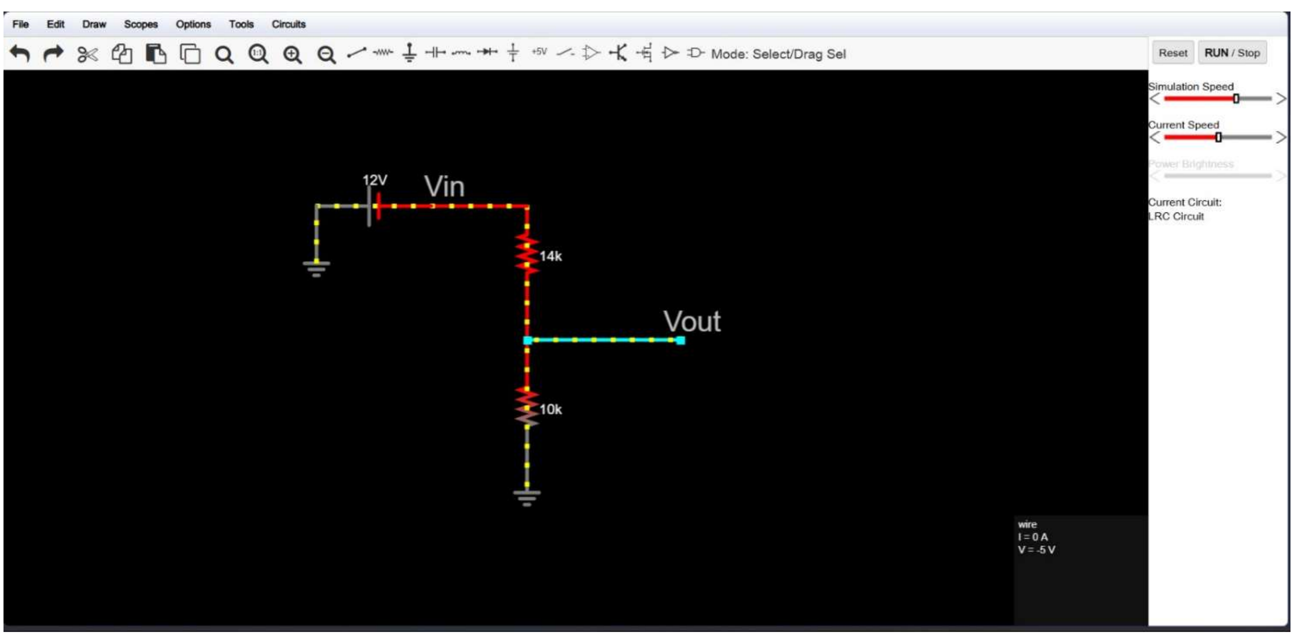
So,
$R_1 = 14K\Omega$
$R_2 = 10K\Omega$

So we will get

$$V_{out} = 12 \cdot \frac{10}{24} = 5V$$

$$P = \frac{(12)^2}{24K} = 6mW$$

Requirmets Satisfied



- **Why have you chosen these specific values for your components?**

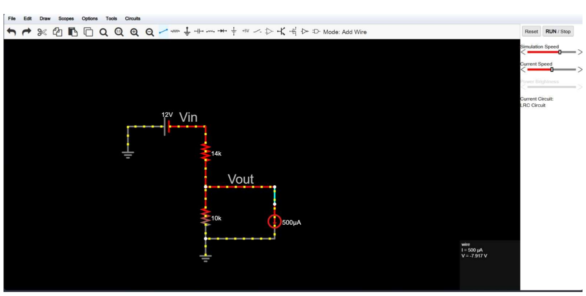
  Because they scale 12V to 5V exactly ,satisfy the 6mW power limit and use only two inexpensive resistors

- **Since you are broke and bought your microcontroller off a cheap Chinese vendor, it turns out the ADC pin is faulty and draws a continuous 0.5mA (not included in the power) of leakage current. What would happen to the output of your circuit? (explain, then simulate to verify your explanation)**

  It only supplies:

$$I = \frac{12}{24k} = 0.5mA$$

If the ADC also draws 0.5mA, the lower resistor effectively has extra current flowing through it, causing the output voltage to drop well below 5V. Which will result in incorrect ADC readings , large measurement error and divider no longer behaves as designed .



- ### What is the maximum acceptable leakage current?

A common design rule is that the ADC leakage current should be much smaller than the divider current to keep errors low.

Using the 10% rule:

$$I_{leak(max)} \approx 0.1 \times I_{divider}$$

With a divider current of 0.5 mA:

$$I_{leak(max)} \approx 0.05 \text{ mA} = 50\ \mu\text{A}$$

So a leakage current of **0.5 mA is far too high**, while **about 50 μA or less** would generally be acceptable for this divider design.